MODEL IDENTIFICATION AND DATA ANALYSIS 2nd module (MIDA2)

Academic Year 2021/2022 – 11/6/2022

Surname	**Name**	**Codice Persona**	**Signature**
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It is not allowed to consult books, notes, lecture notes etc.

Accuracy, clarity and organization of the answers will be evaluated

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1. Given the following impulse response

$$\omega(t) = \frac{4}{3}\left(-\frac{1}{3}\right)^t,$$

answer the following questions:

a. Is the data-generating system stable and strictly proper?
b. Find the transfer function associated with the data-generating system.
c. Identify the system with the N4SID method, assuming that the initial state of the system is equal to zero.
d. Compute the transfer function associated with the identified model and compare it with that of the data-generating system.

a) The properties of the data-generating system can be evaluated by analyzing the impulse response of the system. Over time it can be seen that:

$$\omega(0) = \frac{4}{3}, \qquad \omega(1) = -\frac{4}{9}, \qquad \omega(2) = \frac{4}{27}, \qquad \omega(3) = -\frac{4}{81}$$

Since the initial value of the impulse response is not equal to zero and the absolute value of the samples of the impulse response decreases over time towards zero the data-generating system is NOT strictly proper and it IS asymptotically stable.

b) By applying the definition and the properties of geometric series

$$W(z) = \sum_{t=0}^{\infty} \omega(t) z^{-t} = \frac{4}{3}\sum_{t=0}^{\infty}\left(-\frac{1}{3}z^{-1}\right)^t = \frac{\frac{4}{3}}{1+\frac{1}{3}z^{-1}}$$

c) Since the system is not strictly proper, $\hat{D} \neq 0$. Specifically,

$$\hat{D} = \omega(0) = \frac{4}{3}.$$

The other matrices of the state space model can be obtained by exploiting the standard N4SID procedure.

1. We thus compute the Hankel matrices H_i for increasing values of i, until the rank of the Hankel matrices stop increasing:

   $H_1 = \omega(1) = -\frac{4}{9}$, which has $rank(H_1) = 1$;

   $H_2 = \begin{bmatrix} -\frac{4}{9} & \frac{4}{27} \\ \frac{4}{27} & -\frac{4}{81} \end{bmatrix}$, which has $rank(H_2) = 1$, since the first row can be obtained by multiplying the second one by -3

The order of the model to be identified is thus $n = 1$

2. We construct the matrix R_2 as

$$R_2 = \begin{bmatrix} -\frac{4}{9} & \frac{4}{27} \end{bmatrix}$$

   And O_2 is accordingly obtained to verify

$$H_2 = O_2 R_2 = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}\begin{bmatrix} -\frac{4}{9} & \frac{4}{27} \end{bmatrix} \Rightarrow \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} 1 \\ -\frac{1}{3} \end{bmatrix}$$

3. By applying the formulas: $\hat{F} = O_2(1,:)^{-1} O_2(2,:) = -\frac{1}{3}$, $\hat{G} = R_2(:,1) = -\frac{4}{9}$, $\hat{H} = O_2(1,:) = 1$

d) By applying the formula: $\hat{W}(z) = \hat{H}\left(zI - \hat{F}\right)^{-1}\hat{G} + \hat{H} = \frac{-\frac{4}{9}}{z+\frac{1}{3}} + \frac{4}{3} = \frac{\frac{4}{3}z}{z+\frac{1}{3}} = \frac{\frac{4}{3}}{1+\frac{1}{3}z^{-1}}$. This result confirms the correctness of the previous identification procedure, since the identified and true transfer functions coincide.

2. Given the system described by the following equations

$$\begin{cases} x(t+1) = ax(t) + w(t) \\ y(t) = x(t) + v(t) \end{cases}$$

Where the process noise $w(t) \sim wn(0, q)$ and the measurement noise $v(t) \sim wn(0,1)$ are uncorrelated with the state and among each others, while $a, q \in \mathbb{R}$, with $a \neq 0, q \geq 0$. Answer the following questions:

a. Find the values of a for which the steady-state Kalman filter exists. [Hint: if needed, prove the empirical convergence of $P(t)$ to the solutions of the ARE by selecting a value for a]

b. Is the steady-state prediction error on the state, *i.e.*, $e_x(t) = x(t) - \hat{x}(t|t-1)$, always bounded?

c. If they exist, compute the steady-state Kalman gains for $a = 1$, with $q_1 = \frac{1}{2}$.

a) Note that: $F = a, \quad V_1 = q, \quad H = 1, \quad V_2 = 1, \quad V_{12} = 0.$

Since $V_{12} = 0$:

- When $|a| < 1$, the system is stable and thus the first asymptotic theorem can be applied, and the steady state Kalman filter exists.
- When $|a| \geq 1$ and $q \neq 0$, the system is observable ($O = 1$, with $rank(O) = 1$) and the system is reachable ($R = \Gamma = \sqrt{q}$, with $rank(R) = rank(\Gamma) = 1$) . Therefore, for the second asymptotic theorem the steady state Kalman filter exists.
- When $|a| \geq 1$ and $q = 0$, the system is not reachable and it is not stable. As such, none of the asymptotic theorems can be applied and the existence of the steady state Kalman filter has to be assessed by studying the convergence of the solutions of the DRE to that of the ARE.
  1. DRE: $P(t+1) = a^2P(t) - \frac{a^2P^2(t)}{P(t)+1} = \frac{a^2P(t)}{P(t)+1}$
  2. ARE: $\bar{P}(\bar{P}+1) = a^2\,\bar{P} \Rightarrow \bar{P}(\bar{P}+1-a^2) = 0 \;\Rightarrow\; \bar{P}_1 = 0,\; \bar{P}_2 = a^2 - 1$ (with $\bar{P} \neq -1$)
  3. $P(t) \to \infty \;\Rightarrow P(t+1) \to a^2$; $P(t+1) \to\; \infty$ when $P(t) = -1$; $P(t) = 0$ when $P(t+1) = 0$

     Consider now 3 possible scenarios:
     1) $P(0) = 0$. In this case, the solution of the DRE coincides with that of the ARE at the beginning and, thus, the solution of the DRE will be $P(t) = P(0) = \bar{P}_1 = 0,\; \forall t > 0$.
     2) $P(0) > 0$ and $P(0) < a^2 - 1$. To assess the evolution of the solutions of the DRE in this case, we evaluate the specific case $a = \sqrt{3}$ and we pick $P(0) = 1$. In this case: $P(1) = \frac{3}{2},\; P(2) = \frac{9}{5},\; P(3) = \frac{27}{14}$, ... By recursively computing the solution of the DRE, it can be shown that it slowly converges to the solution $\bar{P}_2$ of the DRE.
     3) $P(0) > 0$ and $P(0) > a^2 - 1$. To assess the evolution of the solutions of the DRE in this case, we evaluate the specific case $a = \sqrt{3}$ and we pick $P(0) = 3$. In this case: $P(1) = \frac{9}{4},\; P(2) = \frac{27}{13},\; P(3) = \frac{81}{40}$, ... By recursively computing the solution of the DRE, it can be shown that it slowly converges to the solution $\bar{P}_2$ of the DRE.

     Therefore, there exists two solutions of the ARE, and the solution of the DRE converges to the non null solution $\bar{P} = a^2 - 1$ whenever $P(0) \neq 0$.

It is thus possible to conclude that the steady-state Kalman filter exists for all $a \neq 0$, with $a \in \mathbb{R}$.

b) The asymptotic theorems guarantee the state prediction error to be bounded whenever $|a| < 1$ or $q \neq 0$. When $|a| \geq 1$ and $q = 0$, we have to check the eigenvalues of

$$F - \bar{K}_i H = a - \frac{a\bar{P}_i}{\bar{P}_i + 1}, \quad i = 1,2$$

Clearly, when considering $\bar{P}_1 = 0$, the filter does not guarantee the error to be asymptotically bounded. On the other hand, when considering $\bar{P}_2$, we have:

$$a - a\frac{(a^2-1)}{a^2} = \frac{1}{a},$$

which is smaller than one in absolute value when $|a| > 1$ and equal to one when $|a| = 1$. In both cases, when $P(0) \neq 0$, the prediction error is bounded, since the dynamic of the prediction error is either simply stable or asymptotically stable, even when $q = 0$.

c) From the formula: $\bar{P}_i = \;\bar{P}_i + q_i - \frac{\bar{P}_i^{\,2}}{\bar{P}_i+1} \Rightarrow\; \bar{P}_1 = 1$. Therefore, the associated steady state Kalman gains are: $\bar{K}_i = \frac{F\,\bar{P}_i}{\bar{P}_i+1} \Rightarrow \bar{K}_1 = \frac{1}{2}$.

3. Consider the following process

$$y(t) = -\frac{1}{3}y(t-1) + u(t-2) + \frac{1}{2}u(t-3) + e(t-2) - \frac{1}{3}e(t-3)$$

with $e \sim wn(0,4)$. Answer the following questions.

a. Are the assumptions for the design of a minimum variance controller satisfied? In case they are not satisfied and if it is possible, manipulate the model of the system for the assumptions of minimum variance control to be satisfied.

b. Design the minimum variance controller for the system.

c. Draw the block-diagram of the closed-loop system.

d. Is the closed loop-system stable? If it is stable, compute the transfer function between the reference $y^o(t)$ and the input $u(t)$.

a) The polynomial characterizing the considered ARMAX process are:

$$A(z) = 1 + \frac{1}{3}z^{-1} \quad B(z) = 1 + \frac{1}{2}z^{-1} \quad C(z) = z^{-2} - \frac{1}{3}z^{-3}$$

with the process equivalently written as:

$$y(t) = \frac{1 + \frac{1}{2}z^{-1}}{1 + \frac{1}{3}z^{-1}}u(t-2) + \frac{z^{-2} - \frac{1}{3}z^{-3}}{1 + \frac{1}{3}z^{-1}}e(t)$$

While the stochastic part of the process is not in canonical form, all the other assumptions for the design of minimum variance controllers are satisfied. We can bring this part of the system in canonical form by simply defining $\eta(t) = e(t-2)$, with $\eta(t) \sim wn(0,4)$, through which

$$y(t) = \frac{1 + \frac{1}{2}z^{-1}}{1 + \frac{1}{3}z^{-1}}u(t-2) + \frac{1 - \frac{1}{3}z^{-1}}{1 + \frac{1}{3}z^{-1}}\eta(t).$$

b) By considering the equivalent representation of the system, we can design the minimum variance control. Since $k = 2$, we have to consider the 2-step ahead predictor of the output, which results in:

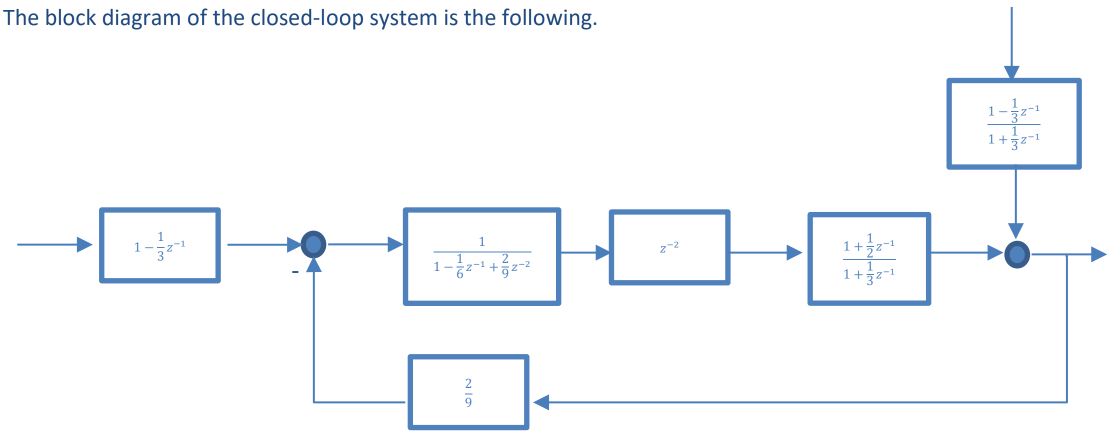
$$E(z) = 1 - \frac{2}{3}z^{-1}, \quad \tilde{R}(z) = \frac{2}{9}$$

By applying the definition, we thus obtain:

$$u(t) = \frac{1}{B(z)E(z)}\left(C(z)y^o(t) - \tilde{R}(z)y(t)\right) = \frac{1}{\left(1 + \frac{1}{2}z^{-1}\right)\left(1 - \frac{2}{3}z^{-1}\right)}\left[\left(1 - \frac{1}{3}z^{-1}\right)y^o(t) + \frac{2}{9}y(t)\right] \Rightarrow u(t)$$

$$= \frac{1}{6}u(t-1) + \frac{1}{3}u(t-2) + y^o(t) + \frac{1}{3}y^o(t-1) - \frac{2}{9}y(t).$$

c) The block diagram of the closed-loop system is the following.



d) To check the stability of the system, we have to show that the roots of the characteristic polynomial of the closed-loop system $\chi(z) = B(z)C(z)$. Since the roots of both $B(z)$ and $C(z)$ are inside the unit circle, the stability of the closed-loop system is guaranteed. Therefore, we compute the transfer function between $y^o(t)$ and $u(t)$, which is given by:

$$W_{y^o u}(z) = \frac{A(z)}{B(z)} = \frac{1 + \frac{1}{3}z^{-1}}{1 + \frac{1}{2}z^{-1}}$$

4. Describe the procedure to estimate a sinusoidal signal (estimation of its amplitude and phase) from a noisy one, assuming that the frequency (Ω) of the sinusoid is known

See lecture notes AY 2021-2022, chapter 2

4. Describe the equations of the state-space method for the discretization of a continuous-time dynamical linear system;

   Discuss the transformation of the poles and zeros of the system from continuous to discrete time.

See lecture notes AY 2021-2022, chapter 6